COMPLETE PROTEIN–LIGAND MD PIPELINE (FROM SCRATCH)

Initial folder

clean.pdb
protein_only.pdb
ligand.gro
ligand.itp
ions.mdp
minim.mdp
nvt.mdp
npt.mdp
md.mdp
topol_Protein_chain_A.itp
posre_Protein_chain_A.itp

STEP 1. Generate protein topology

gmx pdb2gmx -f protein_only.pdb -o protein.gro -water spce -ignh

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STEP 2. Insert ligand into protein

gmx insert-molecules -f protein.gro -ci ligand.gro -o complex.gro -nmol 1

STEP 3. Define simulation box

gmx editconf -f complex.gro -o complex_box.gro -c -d 1.0 -bt cubic

1.0 nm is standard. Smaller is risky.

STEP 4. Solvate system

gmx solvate -cp complex_box.gro -cs spc216.gro -o complex_solv.gro -p topol.top

This updates solvent count automatically.

STEP 5. Edit topology, THIS IS NOT OPTIONAL

Add the ligand.itp definitions in topol.top

```
98840
98841   [ molecules ]
98842   ; Compound        #mols
98843   Protein_chain_A     1
98844   UNK                 1
98845   SOL            237398
98846   CL                  3
98847
```

Add "UNK 1" in topol.top near the end of the file
"UNK" is the name of the ligand.
check the same in ligand.itp
In this case ligand.itp had:

```
37    opls_818  C818    12.0110    0
38  [ moleculetype ]
39  ; Name            nrexcl
40  UNK               3
41  [ atoms ]
42  ;   nr       type  resnr residue
```

```
18  ;
19
20  ; Include forcefield parameters
21  #include "oplsaa.ff/forcefield.itp"
22  #include "ligand.itp"
23
```

Add "ligand.itp" file below the existing "#include" definitions.

5.1 Include ligand.itp

Open `topol.top`

Add **below forcefield include**:

#include "ligand.itp"

5.2 Add ligand to [molecules] section

At the bottom:

[molecules]
Protein_chain_A 1
UNK 1
SOL XXXX

- `UNK` must exactly match ligand name in `ligand.itp`
- If this is wrong, simulation is invalid

STEP 6. Energy minimization

Preprocess

>gmx grompp -f minim.mdp -c complex_solv.gro -p topol.top -o em.tpr -maxwarn 1

Run

>gmx mdrun -v -deffnm em

Check potential energy

>gmx energy -f em.edr -o em_potential.xvg
select: Potential

>xmgrace em_potential.xvg

Energy must smoothly decrease.

STEP 7. Create index file (CRITICAL FOR COMPLEX)

>gmx make_ndx -f em.gro -o index.ndx

Inside prompt:

>1 | 13
>q

This creates the Protein **+ Ligand group**.

STEP 8. Visualize EM result

>gmx trjconv -s em.tpr -f em.gro -n index.ndx -o em_centered.gro -center -pbc mol

Selections:

- Center: Protein_UNK
- Output: System

>vmd em_centered.gro

Visually confirm ligand is bound.

STEP 9. NVT equilibration

Preprocess

>gmx grompp -f nvt.mdp -c em.gro -r em.gro -p topol.top -o nvt.tpr -maxwarn 1

Run

>gmx mdrun -v -deffnm nvt

Temperature check

>gmx energy -f nvt.edr -o nvt_temperature.xvg
select: Temperature

>xmgrace nvt_temperature.xvg

Must stabilize around target temperature.

STEP 10. NVT visualization

>gmx trjconv -s nvt.tpr -f nvt.trr -n index.ndx -o nvt_center.xtc -center -pbc mol -ur compact

>vmd nvt.gro nvt_center.xtc

Check ligand stability again.

STEP 11. NPT equilibration

Preprocess

>gmx grompp -f npt.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -o npt.tpr -maxwarn 1

Run

>gmx mdrun -v -deffnm npt

Pressure

>gmx energy -f npt.edr -o npt_pressure.xvg
select: Pressure

Density

>gmx energy -f npt.edr -o npt_density.xvg
select: Density

Density should settle near 1000 kg/m³.

STEP 12. Production MD

Preprocess

>gmx grompp -f md.mdp -c npt.gro -t npt.cpt -p topol.top -o md.tpr -maxwarn 1

Run

>gmx mdrun -v -deffnm md

STEP 13. Clean trajectory (MANDATORY)

>printf "4\n0\n" | gmx trjconv -s md.tpr -f md.xtc -o md_noPBC.xtc -center -pbc mol -ur compact

>printf "4\n0\n" | gmx trjconv -s md.tpr -f md_noPBC.xtc -o md_fit.xtc -fit rot+trans

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STEP 14. RMSD analysis (protein backbone)

>printf "4\n4\n" | gmx rms -s md.tpr -f md_fit.xtc -n index.ndx -tu ns -o rmsd_backbone.xvg

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STEP 15. Visualize PRODUCTION MD trajectory

You already created the correct trajectory: `md_fit.xtc`.

Load in VMD (correct way)

>vmd md.gro md_fit.xtc

If you want center-of-mass smooth motion:

>vmd md.gro md_noPBC.xtc

STEP 16. Plot RMSD properly (no hand waving)

View RMSD

>xmgrace rmsd_backbone.xvg

What a GOOD RMSD looks like

- Rises initially (equilibration)
- Plateaus
- No continuous upward drift

If RMSD keeps increasing linearly, the system is unstable.

STEP 17. Overlay RMSD for presentation

If you want RMSD in **nanoseconds**:

>gmx rms -s md.tpr -f md_fit.xtc -n index.ndx -o rmsd_ns.xvg -tu ns

Open both:

>xmgrace rmsd_backbone.xvg rmsd_ns.xvg

Pick the cleaner one for paper.

STEP 18. Ligand stability visualization (not optional)

Ligand RMSD relative to protein

>gmx rms -s md.tpr -f md_fit.xtc -n index.ndx -o rmsd_ligand.xvg

Selections:

- Reference: Protein
- RMSD group: UNK

This answers the real question: **does ligand stay put**.

Plot:

>xmgrace rmsd_ligand.xvg

STEP 19. Distance based sanity check

Distance between ligand and protein COM:

>gmx distance -s md.tpr -f md_fit.xtc \
-select 'com of group Protein plus com of group UNK' \
-oall lig_prot_dist.xvg

Plot:

>xmgrace lig_prot_dist.xvg

Flat = stable binding
Spike = ligand drift

Reviewers love this.

STEP 20. Make a clean MD movie (for sanity + presentation)

Create movie friendly trajectory

gmx trjconv -s md.tpr -f md_fit.xtc -o md_movie.xtc -skip 10
>

Load:

>vmd md.gro md_movie.xtc

Then:

- Representation: Cartoon + Licorice for ligand
- Color protein by chain
- Color ligand by element

STEP 21. Optional but strong plots

RMSF (residue flexibility)

>gmx rmsf -s md.tpr -f md_fit.xtc -o rmsf_ca.xvg -res

Radius of gyration

>gmx gyrate -s md.tpr -f md_fit.xtc -o gyration.xvg
